

ABHISHEK CHAURASIYA

Kanpur, India | +91 9793584982 | abhiabckchaurasiya@gmail.com
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Career Summary

Software engineer who designs and ships **production-live** systems in **Python** — across **applied AI**, **backend APIs**, and self-hosted **DevOps**. Built natural-language **semantic image search** (**CLIP embeddings**, **vector retrieval**), a self-hosted **S3-compatible object store**, an **automated deployment platform**, and a backend scaled to **2,000 concurrent users at 0% errors**. Fast, independent builder who debugs to root cause and owns projects end-to-end — **Top 1.5% on LeetCode**; paid freelance delivery.

Education

Pranveer Singh Institute of Technology (PSIT), Kanpur 2023 – 2027
B.Tech, Computer Science (Data Science Specialisation) | CGPA **8.078**

K. R. Education Center, Kanpur 2021 – 2022
Intermediate (Class XII) | 87.66%

Technical Skills

AI / ML: CLIP / sentence-transformers (image & text **embeddings**), **semantic** / **vector search** (cosine similarity, top-k retrieval), zero-shot classification, perceptual hashing (pHash), NumPy.

Languages: Python (Proficient), SQL, JavaScript, TypeScript, C, C++, Java; Data Structures & Algorithms.

Backend & APIs: FastAPI, Django, Django REST Framework, RESTful API design, JSON, WebSocket APIs, JWT authentication, async I/O (**asyncio**), OOP & design patterns, Redis / Valkey.

Databases & Storage: PostgreSQL (JSONB, **pg_trgm**, job queues), MySQL, SQLite, S3 protocol / AWS SigV4 (boto3), multipart uploads.

DevOps & Tools: Docker, Nginx, Linux server administration, Cloudflare Tunnels, Git & GitHub, Postman, VS Code, AI-assisted development (Claude Code, Copilot).

Projects & Engineering Work

PicLocker — Natural-Language Semantic Photo Search May – Jun 2026

Python / CLIP / sentence-transformers / NumPy / SQLite / boto3 / AWS SigV4

- Built **natural-language image search** (knowledge retrieval): encodes photos into **512-dim multimodal CLIP embeddings** and ranks free-text queries by **cosine-similarity nearest-neighbor** (kNN) over an in-RAM, L2-normalised matrix — millisecond top-k, with a tuned threshold so out-of-library queries return empty, not nearest-noise.
- Delivered **zero-shot semantic tagging** (no training, no labels) through CLIP's shared image-text **embedding space**; persisted embeddings as **float32 BLOBs** (~2 KB/photo, ~20 MB index per 10k photos).

PersonalS3 — Self-Hosted S3-Compatible Object Store & Media Platform May – Jun 2026

Python (FFmpeg worker) / PostgreSQL / Valkey / Docker / [Live](#)

- Architected and operate a **production-live**, self-hosted S3-compatible object store and HLS media pipeline across **10 Docker services** on one Linux server (the same store PicLocker indexes), exposed globally via Cloudflare Tunnel.
- Authored end-to-end the **Python data-processing pipeline** that drains a Postgres job queue via **FOR UPDATE SKIP LOCKED** and runs hardware-accelerated FFmpeg transcoding; shipped the Go API + Next.js dashboard via **AI-assisted development** against detailed specs, then reviewed and debugged the output.

URL Shortener — Production-Grade Backend Service Jan – Apr 2026

FastAPI / Redis / PostgreSQL / Docker / Locust / [GitHub](#)

- Scaled a **FastAPI** backend from 200 to **2,000 concurrent users at 0% error rate** by re-architecting the **system design** to Redis-as-source-of-truth (removing PostgreSQL pool exhaustion from the request path); sustained a **load-tested 340 RPS** (Locust) via a write-behind **concurrency** sidcar, deployed with Docker + Gunicorn/Uvicorn.

gitDeploy — Self-Hosted Deployment Automation Platform 2025 – 2026

FastAPI / WebSocket / Docker / Nginx / React / [Live](#)

- Built a **CI/CD-style** pipeline that automates the full GitHub-to-Docker deploy flow into a single **REST API** call on a **production-live** platform, with **WebSocket** real-time log streaming, dynamic port allocation, and automated Nginx config generation.

Experience

Freelance Full-Stack Backend Developer — Niroskos Jan – Mar 2026

Paid contract / Multilingual Travel-Booking Platform / Remote

- Delivered **3 production Django apps** (**10,000+ lines**, ~30 commits) in **collaboration with a paid client** — an **object-oriented** payments module and a Redis-cached **REST/JSON** checkout, hardened across **15+ locales** by a **500+ line integration test suite** (caught N+1, stale-pricing, and locale-resolution defects).

Achievements & Certifications

- Top 1.5% (536th / 36,461)** in LeetCode Biweekly Contest 177 (all 4 problems solved); Codeforces Pupil (1223); **400+ problems** solved.
- Certifications:** Python for Data Science, AI & Development (IBM); Data Analysis with Python (IBM).